

The Midnight Mango Thief!

A SUPER ENGINEER ADVENTURE STORYBOOK

1. The Mystery of the Orchard Thief

Once upon a time, there was a little boy named **Ayaan**. His grandfather grew the most gorgeous, big, yellow Alphonso mango trees in their backyard orchard in Ratnagiri. The mangoes were sweet, juicy, and absolutely perfect.

But oh no! Every single night at midnight, a sneaky, mischievous little monkey climbed down the branches and took a giant nibble out of the very best fruits! Ayaan could not stay awake all night long to watch the tree.

"Can we use our science superpowers to build a secret electronic sound alarm trap to catch the thief in the act?" Ayaan asked. Let's help him protect the mangoes!

2. The Secret of the Electricity Ants

To make our sound alarm work, we need to build a special path called a **circuit**. Imagine that inside every wire, there are billions of tiny, invisible, energetic **Electricity Ants**! These ants love to run and carry power. But they have one major rule: they will only start marching if they have a perfect, unbroken circular track to walk on!

✗ The Open Circuit

If there is a gap or a broken line in the wire, the ants freeze because they cannot jump across! Everything stays completely quiet.

✓ The Closed Circuit

If we connect the lines perfectly, it becomes a complete loop. The ants can march all the way through the wires, and the alarm sings!

3. Circle Tracks are Everywhere!

We see these hidden ant tracks working all around us every single day:



The Big Metro Train: Think of the busy Metro trains zooming through the city. If the circular track is entirely built and continuous, the train runs smoothly from station to station. If a section of track is removed for repairs, the loop breaks and the train stops instantly!



Festival Rice Lights: During festive celebrations like Diwali or Eid, families hang beautiful twinkling fairy lights along their balconies. If just one tiny wire gets clipped or snapped by a sharp window edge, the entire string of lights goes dark immediately because the ant pathway is open!

4. Meet the Sound Maker: The Buzzer

Every alarm system needs a loud mouth, and ours is called the **piezo buzzer**! This clever little part takes the fast running speed of our marching electricity ants and turns it instantly into clear sound waves. You hear buzzers around you all the time:

- ☐ When a delivery person pushes your apartment doorbell button and it sings Ding-Dong!
- ☐ When the school bell loudly echoes across the hallways to tell you it's time for recess!
- ☐ When the microwave loudly beeps to tell you your food is warm and ready to eat!

5. The Secret Mat Trigger

How will our circuit know when the monkey arrives? We will hide a secret **pressure switch** under a garden doormat! Normally, the switch just hangs out in an open state, holding the wires apart so the circuit stays silent and the ants wait patiently.

But the very second the mango thief steps his furry foot down on our mat, his weight forces the hidden metal pads inside to touch! The gap closes instantly, the track is connected, and the ants start to run!

6. Building the Trap Step-by-Step

Let's put our invention together! Stretch out your magic fingers and get ready to connect:

- ☐ **Connect the Power:** Take the positive red wire from your heavy-duty yellow battery pack, pull it across, and snap it directly onto the red leg of our buzzer. Now the power is locked into position!
- ☐ **Route into the Mat:** Take the black wire from the buzzer and run it directly into our secret garden doormat switch.
- ☐ **Complete the Loop:** Connect the other side of the mat switch back to the black side of your battery pack. The loop is complete, and the trap is set!

Midnight Test Time! Imagine dropping a large, heavy stone right on top of our garden doormat switch. The mat squishes down, the hidden metal pads press together, the gap closes, and the electricity ants complete their march all the way to the buzzer!

BZZZZZZZZ! 

7. Be a Science Detective!

Sometimes, real inventors have to solve puzzles when things don't work right. Let's solve these two common mysteries:

Mystery 1: The Upside-Down Wires

What if the buzzer stays totally quiet? Electricity ants from toy batteries can only march in one specific direction. If the red and black wires are accidentally swapped backwards, the ants get confused and freeze. Flipping them back to the right side fixes the track instantly!

Mystery 2: The Plastic Block Wall

What if a plastic toy block gets stuck in the circuit? Materials like plastic and rubber are **insulators**, which means they act like giant brick walls to electricity ants. To fix it, swap the plastic for a metal safety pin! Metals are **conductors**, acting like smooth, open highways where ants love to sprint!

8. The Secret Safety Oath

Small toy batteries have gentle, friendly power that is perfectly safe for young inventors. But the big wall sockets in your house carry giant, dangerous high-voltage electricity! You must never ever touch wall outlets or stick anything into them. Always ask a grown-up first!

Raise your right hand and say this Safety Pledge aloud:

"I promise to always use safe toy batteries for my inventions. I will never, ever touch big household wall plugs, and I will always ask my parents or teacher first!"

9. Victory & Super Engineer Homework!

Hooray! Ayaan's beautiful mango tree is perfectly safe now, thanks to your incredible engineering skills! The sneaky monkey heard our loud buzzer alarm go BZZZZ and safely ran back into the deep jungle without taking a single bite of fruit!



Your Special Homework: Look around your kitchen and living room today and try to spot one device that uses a buzzer or makes a beep. Draw a picture of it in your notebook to share next time!

OFFICIAL AWARD

Junior Lab Assistant

This certificate confirms that a brilliant young creator has completed the **Midnight Mango Thief Trap**, learned the secret of the Electricity Ants, and promised to keep exploring safe circuits!

Ayaan's Grandfather Orchard Signature & Research Partner Witness